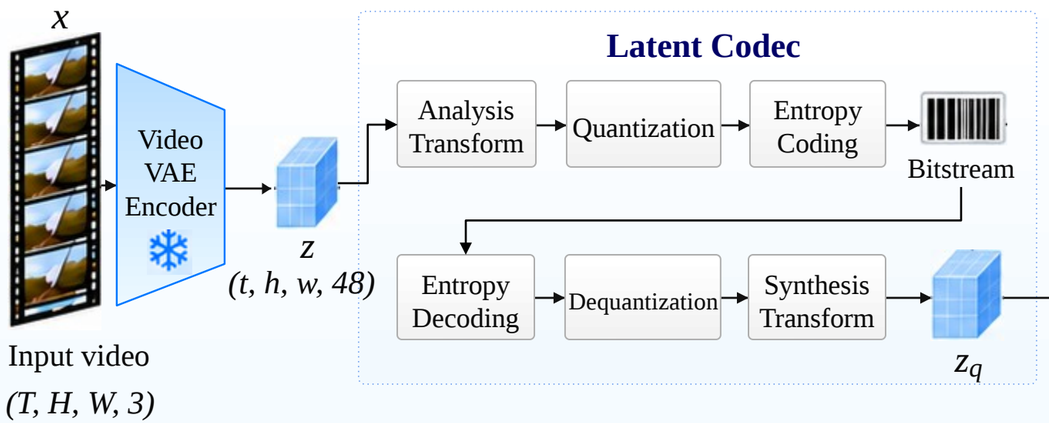
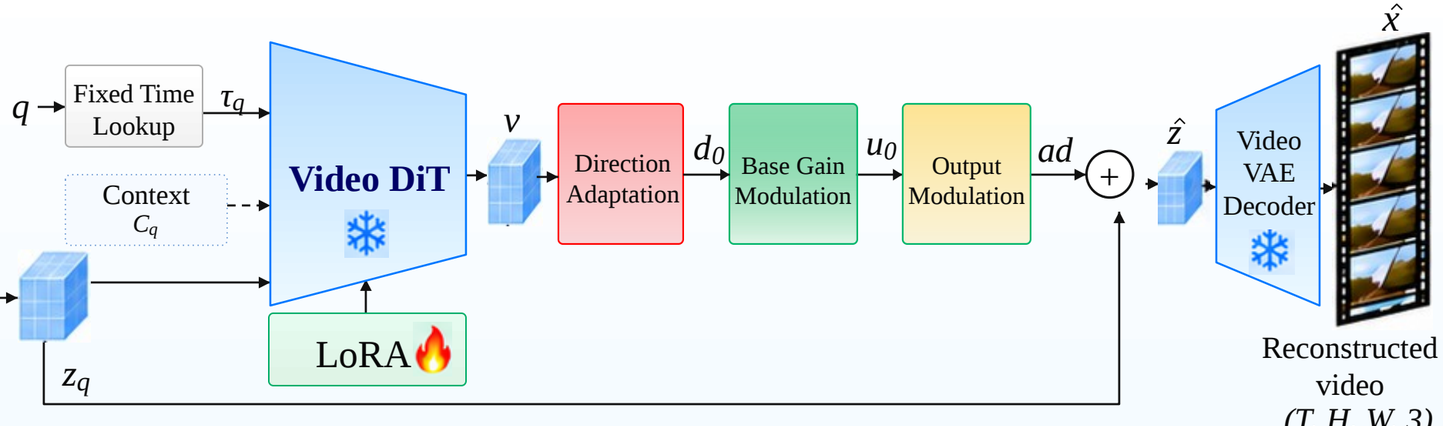
Latent Compression



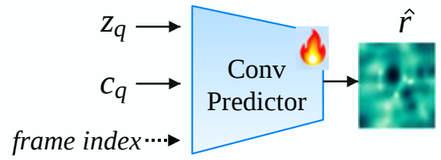
Generative Restoration



Restoration Modules — Details

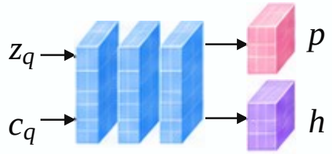
Full-band RMS Estimator

$(z_q, c_q) \rightarrow \hat{r}$



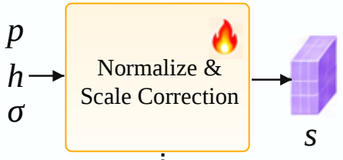
Multi-scale Error Estimator

$(z_q, c_q) \rightarrow (p, h)$



Strength Fusion

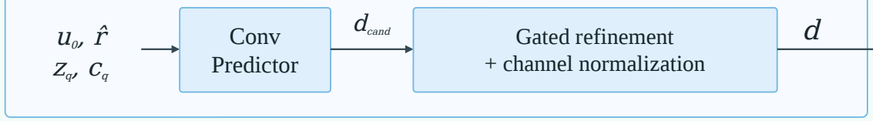
$(p, h, \sigma) \rightarrow s$



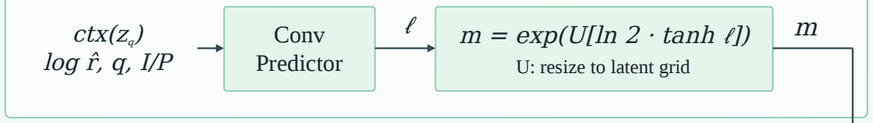
Output Modulation

$(u_0, \hat{r}, z_q, c_q) \rightarrow \Delta z = a \odot d$

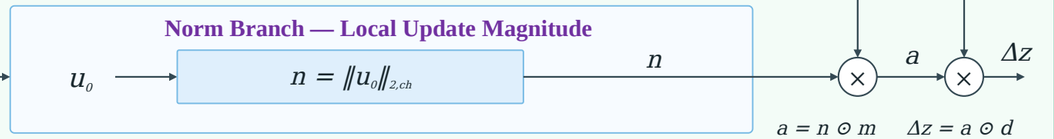
Direction Branch — Gated Angular Refinement



Gain Branch — Local Magnitude Modulation



Norm Branch — Local Update Magnitude



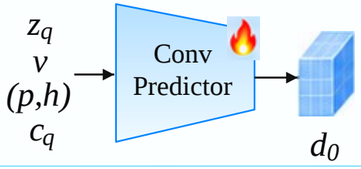
Decoder conditions

$c_q = \{\sigma, w, q, I/P\}$

Entropy scale σ	Regional QP width w
QP set q	Frame role I/P

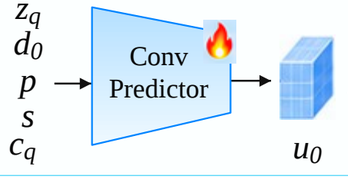
Decoder Adaptation

$(z_q, v, (p,h), c_q) \rightarrow d_0$



Base Gain Modulation

$(z_q, d_0, p, s, c_q) \rightarrow u_0$



$e_0 = Normalize(u_0)$; $d = Normalize[e_0 + \beta_{I/P}(d_{cand} - e_0)]$

d: 48-D unit direction per token n, m, a: scalar maps $\hat{z} = z_q + \Delta z$

Latent feature · Intermediate tensor · Predicted parameter · Scalar map · Data flow · Element-wise addition · Element-wise multiplication · Frozen module · Pretrained module · New module · Trainable module